DTU

Model-based Machine Learning

Probability and Statistics review

Outline

Introduction

Random variable, atom, and event

Joint distribution

Conditional probability

Bayes theorem

Independence

Expectation

Continuous random variables

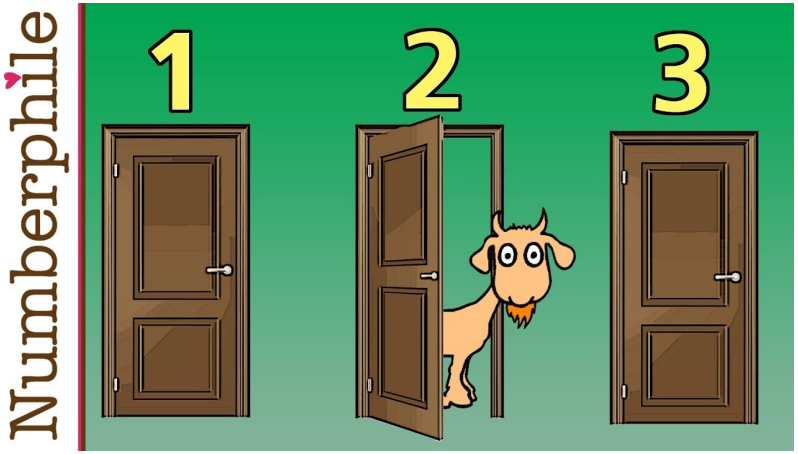
(Based on David MacKay, David Blei,
https://www.cs.princeton.edu/courses/archive/spring12/cos424/pdf/lecture02.pdf)

Teaser

Consider the "Monty Hall problem"

- There are three doors:
  - One has a car (picture)
  - Two have a goat (picture)

  1. Participant chooses one door
  2. Host (*Monty Hall*) opens another door
  3. Host's opened door is always a goat
- Should the participant change his/her choice?



Random variable, atom, and event

- In Algebra a variable, x, is an unknown value
  - E.g. $2x = 4$
  - It can take at most one value at a time
- A **random variable** represents simultaneously a set of values
- Necessary in contexts where we *cannot* determine a unique value
  - Of course, theoretically, it also corresponds to one value...
  - But we can only determine its distribution
  - E.g. $p(5 < X < 10) = 0.5$
- It can be a single value, a vector, a matrix...

Random variable, atom, and event

- Random variables take on values in a sample space
- They can be discrete or continuous
- For example:
  - Coin flip: $\{H, T\}$
  - Height: Positive values $(0, \infty)$
  - Temperature: real values $(-\infty, \infty)$
  - Number of words in a document: Positive integers $\{1, 2, ..., \infty\}$
- We call the values of random variables *atoms*

Random variable, atom, and event

- A *discrete probability distribution* assigns probability to every atom in the sample space
- For example, if X is an (unfair) coin, then
  - $p(X = H) = 0.7$
  - $p(X = T) = 0.3$
- The sum of probabilities of *any* distribution is 1

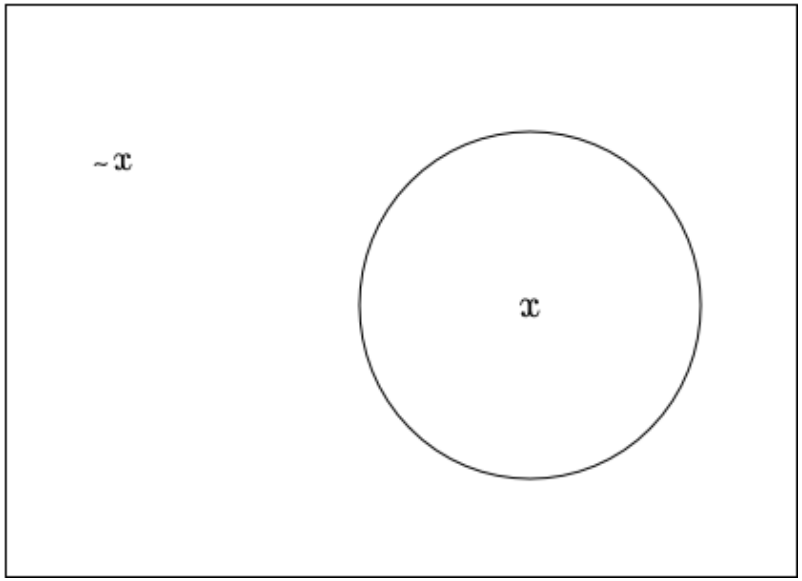
$$\sum_x p(X = x) = 1$$

- And all probabilities have to be greater or equal to 0
- Probabilities of disjunctions are sums over part of the space. E.g., the probability that a die is bigger than 3:

$$p(X > 3) = p(X = 4) + p(X = 5) + p(X = 6)$$

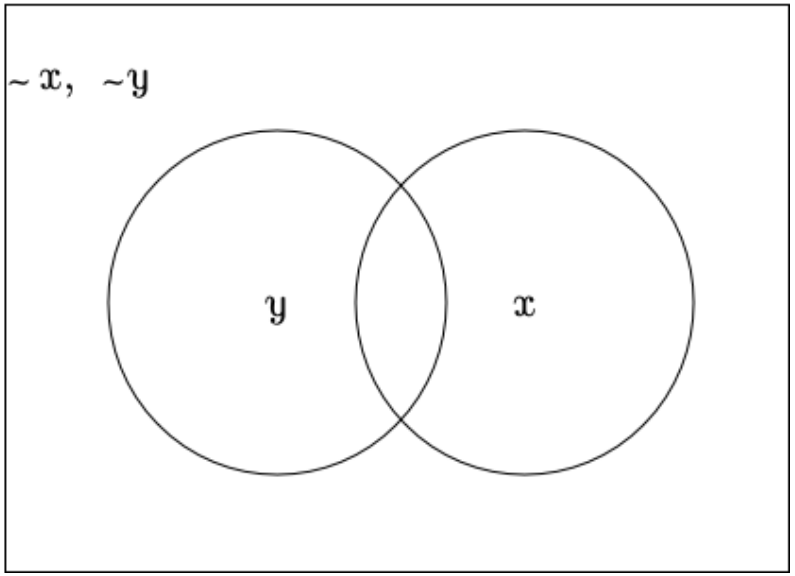
Random variable, atom, and event

- The figure below is helpful to understand these concepts well



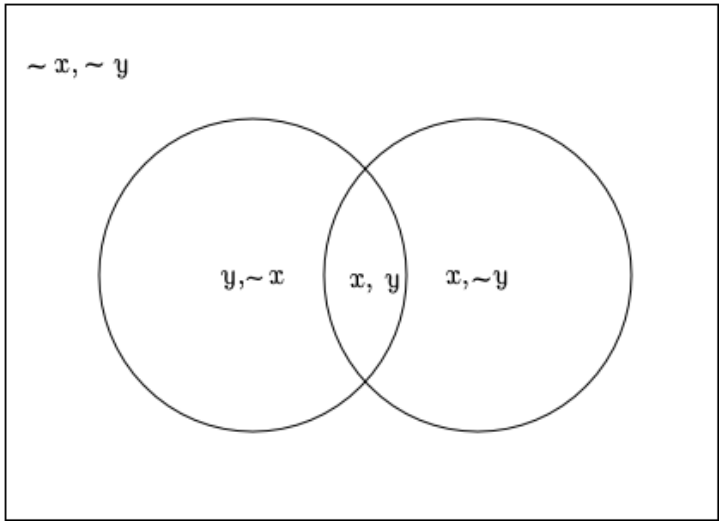
- An *atom* is a point in the box. All atoms together form the *sample space*
- An *event* is a subset of atoms. Two events in the picture are x and $\sim x$
- The probability of an event is the sum of the probabilities of its atoms

- In practice, we often combine many variables/events at the same time



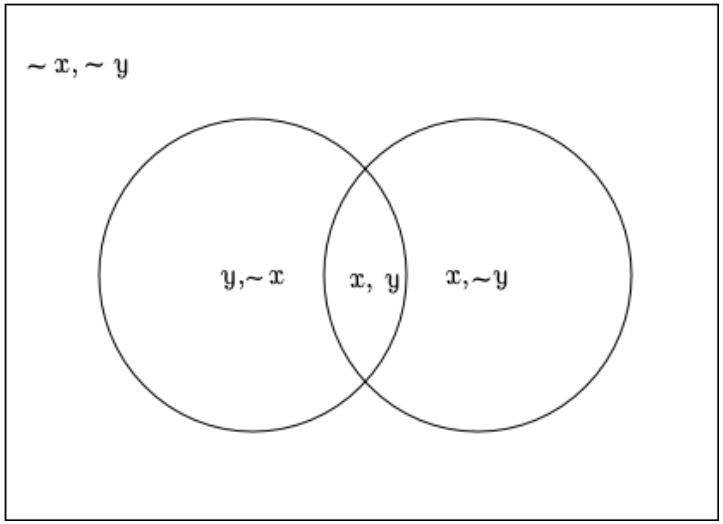
- The **joint distribution** is a distribution over the configuration of all the random variables in the ensemble
  - For the figure, the function $p(X, Y)$ gives the probability of all possible combinations of X and Y
  - Notice that $X \in \{x, \sim x\}$ and $Y \in \{y, \sim y\}$
  - Therefore $X, Y \in \{(x, y), (x, \sim y), (\sim x, y), (\sim x, \sim y)\}$

Joint distribution



- Some useful properties:
  - Union: $p(X \cup Y) = p(X) + p(Y) - p(X, Y)$
  - **Marginalization**: $p(X) = \sum_Y p(X, Y)$
    This property is referred to as the **sum rule of probability!**

Joint distribution



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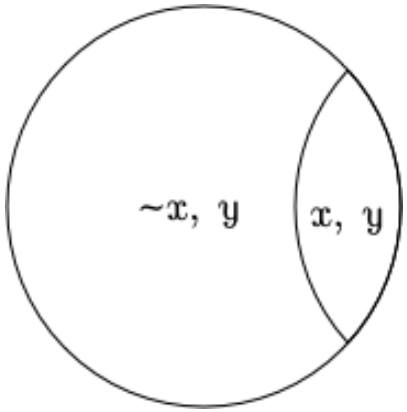
  - **Marginalization**: $p(X) = \sum_Y p(X, Y)$
    This property is referred to as the **sum rule of probability!**
  - Notice that you can have more variables on the conditioning side:
    $p(X|Z, \ldots) = \sum_Y p(X, Y|Z, \ldots)$
  - Or marginalize over more variables: $p(X) = \sum_Y \sum_Z \ldots p(X, Y, Z, \ldots)$

Conditional probability

- What about when we have observed one event, but want to know the probability of another one?
- The **conditional probability** of X given Y is the probability of event X when event Y is known



- So, we only concentrate on the subset of events where the specific value of Y occurs
- In the above figure, we focus on when $Y = y$

$$p(X|Y = y) = \frac{p(X, Y = y)}{p(Y = y)}$$

The chain rule (or product rule)

- Consider the conditional probability rule

$$p(X|Y) = \frac{p(X,Y)}{p(Y)}$$

- It allows us to derive the chain rule, which defines the joint distribution as a product of conditionals:

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- In general, for any set of variables

$$p(X_1, X_2, ..., X_N) = \prod_{n=1}^{N} p(X_n|X_1, X_2, ..., X_{n-1})$$

- For example:

$$p(X,Y,Z) = p(X)\,p(Y|X)\,p(Z|Y,X)$$

Bayes theorem

- Using the chain rule, we can trivially say:

$$p(X|Y)\,p(Y) = p(Y|X)\,p(X)$$

which means that [***Bayes theorem***]:

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$$p(X|Y, Z, \ldots) = \frac{p(Y|X, Z, \ldots)\,p(X|Z, \ldots)}{p(Y|Z, \ldots)}$$

- The Bayes theorem is an important foundation for Bayesian statistics, and particularly for Probabilistic Graphical Models!

Independence

- Random variables are *independent* if knowing about X tells us nothing about Y

$$p(Y|X) = p(Y)$$

- This means that their joint distribution is

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- A few examples:
  - Two lottery numbers that two (unacquainted) people chose. Are these two numbers independent?
  - Two persons, A, and B, start their trip in different parts of town. The transport mode for A is X and for B, it is Y. Are these two choices independent?
  - It's a rainy day. Two accidents happen on different roads of the city. Are these two, independent events?
  - The speeds in adjacent road sections x

Independence

- Example: two coins, C_1, C_2 with $p(H|C_1) = 0.6, p(H|C_2) = 0.2$
  1. Suppose that I randomly choose a number $Z \in \{1, 2\}$ (with equal probability), and take coin C_Z
  2. I flip it twice, with results (X_1, X_2)

  Are X_1 and X_2 independent? What about if I know Z?

Conditional independence

- X and Y are *conditionally independent* given Z

$$p(X|Y,Z) = p(X|Z)$$

- So, we can say that

$$X \perp\!\!\!\perp Y|Z \implies p(X,Y|Z) = p(X|Z)\,p(Y|Z)$$

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- *If we know Z, then knowing about Y tells us nothing about X*

Teaser - Monty Hall

- We can now solve the Monty Hall problem

  X = true location of the car
  Y = door that host opened
  Z = choice of participant

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  we can also say:

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  We know that, $p(X)$, the prior probability of the location is:

$$p(X) = \frac{1}{3}$$

  But what about $p(Y|X,Z)$?

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	$Y = 1$	$Y = 2$	$Y = 3$
$X = 1, Z = 1$	0	0.5	0.5
$X = 1, Z = 2$	0	0	1
$X = 1, Z = 3$	0	1	0
$X = 2, Z = 1$	0	0	1
$X = 2, Z = 2$	0.5	0	0.5
$X = 2, Z = 3$	1	0	0
$X = 3, Z = 1$	0	1	0
$X = 3, Z = 2$	1	0	0
$X = 3, Z = 3$	0.5	0.5	0

Table: $p(Y|X, Z)$

Teaser - Monty Hall

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$X = 2, Z = 2$	0.5	0	0.5
$X = 2, Z = 3$	1	0	0
$X = 3, Z = 1$	0	1	0
$X = 3, Z = 2$	1	0	0
$X = 3, Z = 3$	0.5	0.5	0

Table: $p(Y|X, Z)$

Ok, let's try a scenario. Let's assume that the participant chose door 3 and the host opened door 2. Then our table becomes:

	$Y = 2$
$X = 1, Z = 3$	1
$X = 2, Z = 3$	0
$X = 3, Z = 3$	0.5

Table: $p(Y = 2|X, Z = 3)$

Teaser - Monty Hall

In this scenario, we want to calculate

$$p(X|Y=2, Z=3) \propto p(Y=2|X, Z=3)P(X)$$

and we have

	$Y=2$
$X=1, Z=3$	1
$X=2, Z=3$	0
$X=3, Z=3$	0.5

Table: $p(Y=2|X, Z=3)$

$$P(X) = \frac{1}{3}$$

- Let's just calculate for the two possible cases (X is either in door 1 or 3!):

$$p(Y=2|X=1, Z=3) \times \frac{1}{3} = \frac{1}{3}$$

$$p(Y=2|X=3, Z=3) \times \frac{1}{3} = \frac{1}{2} * \frac{1}{3} = \frac{1}{6}$$

- So we can get our normalizing quantity: $\sum_X p(Y|X, Z)p(X) = \frac{1}{3} + \frac{1}{6} = \frac{1}{2}$

Teaser - Monty Hall

Remember: to calculate the distribution of X (i.e. “where the car probably is"), we need to calculate

$$p(X|Y,Z) = \frac{p(Y|X,Z)p(X)}{\sum_X p(Y|X,Z)p(X)}$$

- If we follow our example, we get

$$p(X|Y=2,Z=3) = \frac{p(Y=2|X,Z=3) \times \frac{1}{3}}{\frac{1}{2}}$$

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$$p(X|Y=2,Z=3) = \frac{p(Y=2|X,Z=3) \times \frac{1}{3}}{\frac{1}{2}}$$

- using the calculations from the previous slide, we have

$$p(X=1|Y=2,Z=3) = \frac{\frac{1}{3}}{\frac{1}{2}} = \frac{2}{3}$$

$$p(X=3|Y=2,Z=3) = \frac{\frac{1}{6}}{\frac{1}{2}} = \frac{1}{3}$$

- By this reasoning, we **always** have $\frac{2}{3}$ chances when we change doors, and keep $\frac{1}{3}$ if we keep it!

Expectation

- The *expected value* of a random variable is the probability-weighted average of all possible values
- In other words, it is the *mean* of the distribution of this random variable

$$\mathbb{E}[X] = \sum_x x\, p(X = x)$$

- More generically (remember the $f(x)$ can be itself a random variable)

$$\mathbb{E}[f(X)] = \sum_x f(x)\, p(X = x)$$

Playtime!

- Open “01 - Probability theory review.ipynb” in Jupyter
- Do Part 1, estimated duration 20 min
- Do Part 2, estimated duration 30 min
- Do Part 3, estimated duration 10 min

Continuous random variables

- We've only used discrete random variables so far (e.g., dice, cards)
- Random variables can be continuous
- We need a density function $p(x)$, which integrates to one.

$$\int_{-\infty}^{\infty} p(x)\, dx = 1$$

- Probabilities are integrals over $p(x)$
- An *event* is thus defined by an interval of possible values of the random variable

$$P(a \leq X \leq b) = \int_{a}^{b} p(x)\, dx$$

- Notice that we use X, x, P, and p!...

Some distributions - Gaussian

- By far, the most common one...
- Two parameters:
  - Mean, μ
  - Standard deviation, σ (or, variance, σ^2)
- $p(x)$ is defined as

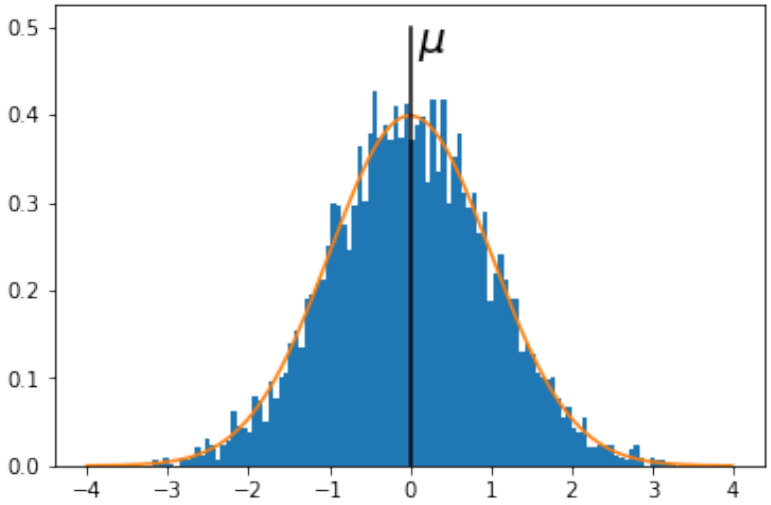
$$p(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)}$$

- Often represented as:

$$p(x) \sim \mathcal{N}(\mu, \sigma^2)$$

Some distributions - Gaussian

- Support is $]-\infty, \infty[$
- Symmetrical



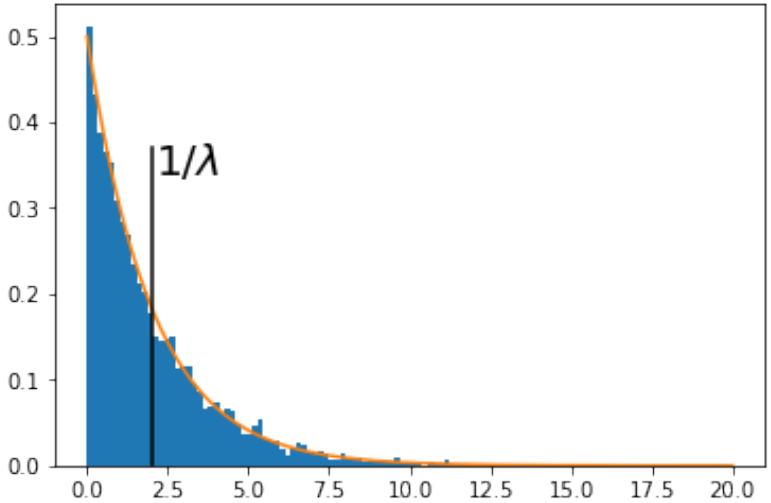
- The Central limit theorem (CLT) establishes that *the distribution of the sampling means approaches a normal distribution as the sample size gets larger, no matter what the shape of the population distribution.*

Some distributions - Exponential

- Exponential distribution, with *rate* λ

$$p(x) = \lambda e^{-\lambda x}$$

- Support is $[0, \infty[$

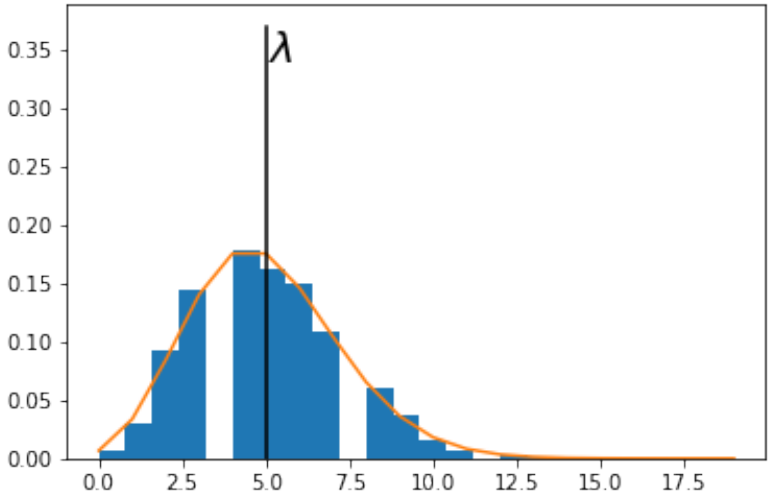


Some distributions - Poisson

- Poisson distribution, with *rate* λ

$$p(k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

- for $k = 0, 1, 2...$
- Pretty common in transportation (e.g. arrival rates)



[1]

[1] In fact, this distribution relates to a discrete random variable, so we include it to emphasize that not only continuous variables can be parameterized as a probability distribution.

Independent and identically distributed random variables (iid)

- Independent
- Identically distributed

Independent and identically distributed random variables (iid)

- Independent
- Identically distributed

If we repeatedly flip the same coin N times and record the outcome, then $X_1, ..., X_N$ are **iid**

- The iid assumption can be extremely useful in data analysis

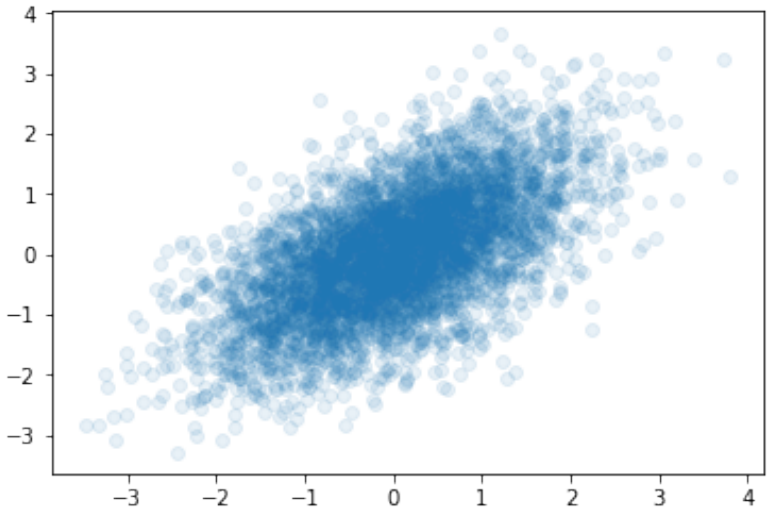
Multivariate distributions

- So far, we've been working with single variable distributions
- Multivariate means it's the same as above, but with more variables at the same time!
- In practice, joint distribution of variables that share a common structure
- In some cases (e.g. Poisson), it is not a trivial problem
- In others (e.g. Gaussian), it is well studied, and extensively applied

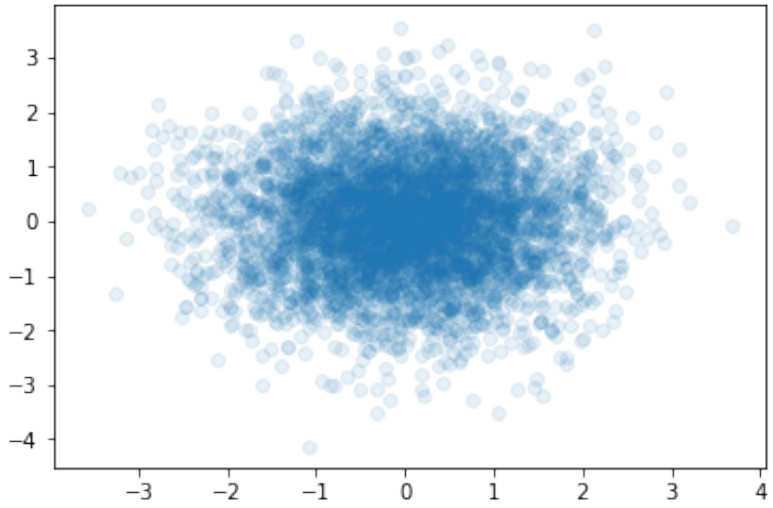
$$p(\mathbf{x}) = \frac{1}{\sqrt{2\pi}|\boldsymbol{\Sigma}|} e^{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})}$$

Multivariate distributions

- Bivariate Gaussian



$$\mathbf{\Sigma} = \begin{bmatrix} 1 & 0.6 \\ 0.6 & 1 \end{bmatrix}$$



$$\mathbf{\Sigma} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

A note on notation

- So far we have been using a rather standard statistics notation
  - X is a random variable and x is atom/event
  - We write e.g. $p(X = x)$
- In the machine learning literature, this notation is typically simplified
  - Lowercase letters, such as x, represent random variables
  - We simply write $p(x)$. Everything else should be clear from the context!
- This allows us to have
  - Bold letters denote vectors (e.g. $\mathbf{x}$, where the ith element is referred as x_i)
  - Matrices are represented by bold uppercase letters such as $\mathbf{X}$
  - Roman letters, such as N, denote constants
- This is the notation that we will adopt from now on!

The likelihood function

- Imagine you have the data. For example:
  - N readings of traffic counts at a certain time, each one called x_i, $i = 1...N$
- You assume it follows some parametric distribution (e.g. Gaussian)
- How do you determine its parameters, Θ?

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- Notice that this is the joint distribution of all **independent** data points!
- In the case of the Gaussian, we should have $\Theta = \{\mu, \sigma\}$
- The likelihood function, $L(\Theta)$, would be

$$L(\Theta) = \prod_{i}^{N} \frac{1}{\sqrt{2\pi}\sigma} e^{\left(-\frac{(x_i-\mu)^2}{2\sigma^2}\right)}$$

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- If you actually *had* the true parameters, the likelihood function would have the maximum value, right?
- So, this becomes an optimization problem:
  - Find the values of Θ that maximize the function $L(\Theta)$

The log-likelihood function

- For practical reasons, we apply a logarithmic transformation to the likelihood function
  - Less prone to numeric error (numerical stability)
  - Computationally faster

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  - Less prone to numeric error (numerical stability)
  - Computationally faster
- In the case of the Gaussian distribution, the log likelihood becomes:

$$-\frac{N}{2}(\log(2\pi) + \log(\sigma^2)) - \frac{1}{2\sigma^2}\sum_i (x_i - \mu)^2$$

Maximum likelihood estimate (MLE)

- The maximum likelihood estimate is the value of the parameter that maximizes the log likelihood (equivalently, the likelihood)
- In the case of the Gaussian, the MLE corresponds to:

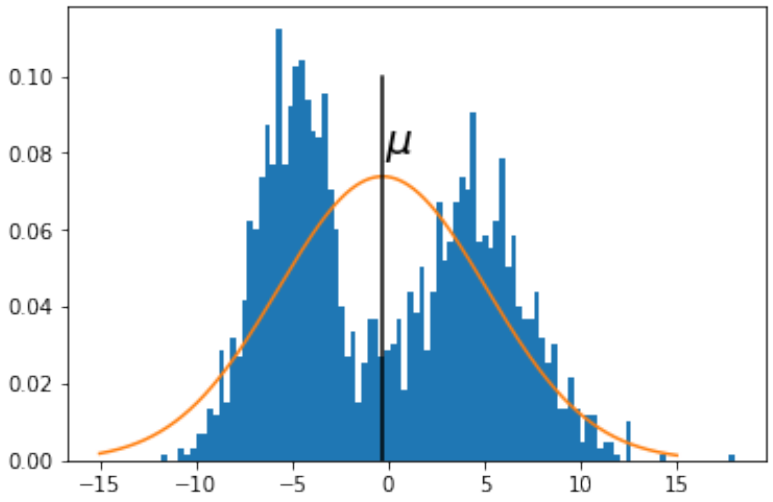
$$\hat{\mu} = \frac{\sum_{i=1}^{N} x_i}{N}, \quad \text{i.e. the } \textit{sample mean}$$

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^{N} (x_i - \hat{\mu})^2}{N}, \quad \text{i.e. the } \textit{sample variance}$$

Maximum likelihood estimate (MLE)

DISCLAIMER:

- The fact that you get a MLE doesn't mean you found a good model!



- You need to know your data...

Playtime!

- Open "01 - Probability theory review.ipynb"
- Do part 4. Est. time is 15 min
- Do part 5. Est. time is 30 min